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Supported iron catalysts for slurry phase Fischer–Tropsch synthesis

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Abstract

The present study was undertaken to investigate reduction and catalytic behavior of silica and alumina-supported iron Fischer–Tropsch (F–T) catalysts promoted with potassium and copper. It was found that the alumina (Vista Catapal B) inhibits reduction of iron, whereas the reduction behavior of the silica (Davison Grade 952) supported catalyst was similar to that of the two precipitated iron catalysts synthesized in our laboratory. Supported and precipitated ($Fe/Cu/K/SiO_2$) catalysts were tested in a stirred tank slurry reactor. Initial activity of the silica-supported catalyst, measured by a value of an apparent first-order reaction rate constant, was 20–40% higher than that of the precipitated iron catalysts, whereas the activity of the alumina-supported catalyst was about 50% less than that of the silica-supported catalyst. Silica-supported catalyst had higher gaseous selectivity and lower olefin content than the alumina-supported catalyst and precipitated catalysts. These results were explained in terms of interactions between potassium with supports, resulting in reduction of the potassium promotion effectiveness. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Fischer–Tropsch synthesis; Supported and precipitated iron catalysts; Slurry reactor; TPR characterization; Isothermal reduction behavior

1. Introduction

Fischer–Tropsch (F–T) hydrocarbon synthesis was first developed and practiced in Germany during the 1930s and 1940s, using fixed bed reactors and cobalt based catalysts. Subsequently, Sasol (South Africa) commercialized the process in mid 1950s in tubular fixed bed (TFB) and circulating fluidized bed (CFB) reactors, using potassium promoted iron catalysts [1]. TFB reactors are used for production of high molecular weight hydrocarbons (waxes) and diesel fuel, whereas CFB reactors are used for production of gasoline and alpha-olefins. Each of these two types of reactors has a rather narrow range of operating conditions with respect to fresh feed composition and the reaction temperature. Both reactors are not suitable for direct processing of synthesis gas produced in modern coal gasifiers, where hydrogen to carbon monoxide molar ratio is 0.5–0.7. Slurry processing provides the ability to more readily remove the heat of reaction, minimizing temperature rise across the reactor and eliminating localized hot spots. As a result of the improved temperature control, yield losses to methane are reduced and catalyst deactivation due to coking is decreased. This, in turn, allows much higher conversions per pass, minimizing synthesis gas recycle, and offers the potential to operate with CO-rich synthesis gas feeds without the need for prior

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water–gas shift [2]. Capital cost of a slurry bubble column reactor (SBCR) is 20–40% lower than that of a fixed bed reactor [3–5]. This type of reactor has much greater flexibility than the fixed bed and fluidized bed reactors, and can operate in either gasoline or wax mode of operation [6]. A semi-commercial scale SBCR (capacity of 2500 barrels per day) has been in operation at Sasol since 1993 [5,7].

Texas A&M University (TAMU) has been working on development of improved iron F–T catalysts for slurry phase process since 1987. Some of the precipitated iron catalysts (Fe/Cu/K/SiO_2) synthesized and tested at TAMU have proven to be more active than iron catalysts employed in the two most successful slurry phase tests conducted by Mobil [6] and Rheinpreussen [8]. In stirred tank slurry reactor tests at TAMU [9–11] the yield of liquid and wax hydrocarbon products (C_5+ hydrocarbons) was 80–89%, and the Anderson–Schulz–Flory chain growth parameter was 0.92–0.95 (high-alpha catalysts) at syngas conversion of about 80%. Even though the performance of these catalysts was excellent in terms of their activity, selectivity and stability, there is a concern that precipitated catalysts may be structurally too weak for SBCR operation. Problems with catalyst/wax filtration were encountered during F–T demonstration run at LaPorte, Texas in a SBCR using a catalyst prepared by United Catalysts Inc. [12]. The filtering system was plugged after 1 day of operation, and the external catalyst/wax separation in a settling tank was inefficient, resulting in gradual loss of catalyst from the reactor. Both problems were attributed to break up of the original catalyst particles to fine particles. Also, Jager and Espinoza [5] stated that the solid/wax separation was a major developmental challenge during Sasol's work on slurry F–T process. This problem was alleviated through preparation of micro-spherical particles by spray drying followed by calcination.

The use of preformed support materials is a potential way to alleviate the catalyst particle breakage (attrition) problem during slurry phase operation. Catalysts prepared by impregnation of support will have attrition properties of the support material, and the latter may have superior attrition resistance (less particle breakage) than a precipitated iron catalyst [13]. Cobalt F–T catalysts are prepared by impregnation of support materials (alumina, silica and/or titania), and reports from relatively short slurry reactor tests indicate that these particles do not break up readily and their breakage does not generate a large fraction of fine particles [13,14]. Previous studies with supported iron catalysts are rather limited [1,13,15–19] and only a few of them dealt with the effect of support type and metal content on catalyst activity and selectivity at elevated pressures (i.e. above the atmospheric pressure). Anderson [17] stated that iron catalysts with high support-to-metal ratios have been ineffective as F–T catalysts, but did not provide any experimental data to corroborate this statement. On the basis of studies conducted at Sasol, Dry [1] stated that impregnation of high surface area carriers (silica, alumina or silica–alumina) with iron nitrate yields inferior catalysts in comparison to precipitated iron catalysts. Lower activity and wax selectivity of the supported catalysts were ascribed to formation of chemical compounds between the alkali promoter (K_2O) and carriers, which results in reduction of alkali promotional effect. However, no experimental data were presented to illustrate and quantify differences in performance between supported and precipitated iron catalysts. Recently, O'Brien et al. [13] evaluated several supported iron F–T catalysts in a slurry reactor and compared their attrition properties, activity and selectivity behavior to that of a precipitated iron catalyst. They found that the precipitated catalyst is more active and produces less methane than the supported catalysts, which is in qualitative agreement with Sasol's experiments in fixed bed reactors [1]. The attrition properties of the supported catalysts were superior to that of the precipitated catalyst, and O'Brien et al. [13] concluded that the supported iron catalysts merit consideration for use in slurry phase F–T synthesis.

Due to limited amount of data available on performance of supported iron catalysts and their potential for use in slurry reactors, we have initiated research on development of supported iron catalysts for slurry reactor applications. Here, we report results from studies of reduction behavior (under both temperature-programmed and isothermal conditions) and catalytic performance during F–T synthesis, in a stirred tank slurry reactor, of two supported iron catalysts promoted with potassium and copper. The same types of results obtained with two highly active and selective precipitated iron catalysts, synthesized at TAMU, are included for comparison and assessment of feasibility of using supported iron catalysts for F–T synthesis in slurry reactors.

2. Experimental

2.1. Catalyst synthesis procedure

Commercial support materials (silica—Davison Grade 952; and boehmite alumina—Vista Catapal B) were first sieved to 140–325 mesh (45–106 μm range) and calcined at 500 °C in air for 5 h prior to impregnation by incipient wetness method. Catalysts were prepared by co-impregnation with aqueous solutions containing desired amounts of ferric nitrate, copper nitrate and potassium bicarbonate in successive steps. Three impregnation steps were required for preparation of the silica-supported catalyst with the desired iron loading, whereas seven impregnation steps were needed for preparation of the alumina-supported catalyst (due to a smaller pore volume of the alumina support). After each impregnation step the sample was vacuum dried at 100 °C for about 2 h. After the final impregnation and drying for 12 h in vacuum, the catalyst was calcined in air at 300 °C for 5 h. Nominal compositions (on mass basis) of synthesized catalysts were 100Fe/5Cu/6K/139SiO_2 and 100Fe/5Cu/9K/139Al_2O_3. The corresponding weight percent of iron (as metal) in the prepared catalysts is approximately 33.8%. This metal loading is high for supported catalysts, but is necessary in order to achieve high reactor productivity. Procedure used for synthesis of precipitated iron catalysts with nominal compositions: 100Fe/3Cu/4K/16SiO_2 (59.7% Fe) and 100Fe/5Cu/6K/24SiO_2 (55.4% Fe) has been described elsewhere [20,21].

2.2. Catalyst characterization equipment and methods

BET surface area and pore volume measurements were obtained by nitrogen physisorption at 77 K using Micromeritics Digisorb 2600 system. Samples (∼1–2 g) were degassed at 150 °C for 12 h prior to each measurement. Surface area only was also determined using Pulse Chemisorb 2705 instrument (Micromeritics Inc.). Catalyst samples were outgassed in a flow of nitrogen (∼30 cm^3/min) at 200 °C for 3–12 h prior to each measurement.

Temperature-programmed reduction (TPR) studies were performed using 5% H_2/95% N_2 as reductant. In a typical TPR experiment about 10–20 mg of catalyst was packed in a quartz reactor and purged with helium to remove the moisture from the catalyst sample. Then the catalyst sample was heated in a flow of 5% H_2/95% N_2 (flow rate of 40 cm^3/min) from room temperature to 800–900 °C at a heating rate of 20 °C/min. Hydrogen consumption was monitored by change in thermal conductivity of the effluent gas stream. A dry ice/acetone bath was used to remove water formed during hydrogen reductions. In order to quantify the degree of reduction, CuO standard was used for calibration of the peak areas. All experiments were conducted in Pulse Chemisorb 2705 unit (Micromeritics Inc.) equipped with thermal conductivity detector and temperature programmable furnace.

Isothermal reduction in thermogravimetric analysis (TGA) experiments was conducted using approximately 20 mg catalyst samples in a simultaneous TGA/DTA apparatus (TA Instruments, Model SDT 2960). The catalyst sample was purged with helium (40 cm^3/min) and temperature was ramped at a rate of 5 °C/min from room temperature to a desired reduction temperature. Then the helium flow was switched to a reductant (hydrogen or carbon monoxide) at 100 cm^3/min, and the temperature was maintained at a constant value up to 9 h.

2.3. Reactor system and operating procedures

Experiments were conducted in a 1 dm^3 stirred tank slurry reactor (Autoclave Engineers). The feed gas flow rate was adjusted with a mass flow controller and passed through a series of oxygen removal, alumina and activated charcoal traps to remove trace impurities. After leaving the reactor, the exit gas passed through a series of high and low (ambient) pressure traps to condense liquid products. High molecular weight hydrocarbons (wax), withdrawn from a slurry reactor through a porous cylindrical sintered metal filter, and liquid products, collected in the high and low pressure traps, were analyzed by capillary gas chromatography. The reactants and non-condensable products leaving the ice traps were analyzed on an on-line GC (Carle AGC 400) with multiple columns using both flame ionization and thermal conductivity detectors. Further details on the experimental set up, operating procedures and product quantification can be found elsewhere [9,22,23].

Table 1
Pretreatment conditions and test designations

Catalyst	Test ID	Reducing gas	Total flow rate[a] (cm^3/min)
100Fe/5Cu/6K/139SiO_2	SB-1327	CO/He = 1/8	7000
100Fe/5Cu/9K/139Al_2O_3	SB-3007	CO/He = 1/5	4000
100Fe/3Cu/4K/16SiO_2	SA-0946	CO	750
100Fe/5Cu/6K/24SiO_2	SB-2537	CO/He = 1/10	5500

Pretreatment conditions: 280 °C, 0.8 MPa, 8 h.
[a] Flow rate at standard temperature and pressure (0 °C and 1 bar).

Calcined catalysts of particle size less than 270 mesh (53 μm) were loaded into the reactor and activated in situ with either pure CO or CO diluted with helium at 280 °C, 0.8 MPa, 3 l (STP) (CO)/g_{cat} h for 8 h (Table 1). Durasyn 164-oil (a hydrogenated 1-decene homopolymer liquid, $\sim C_{30}$) was used as a start-up fluid and catalyst loading was about 5–7 wt.% in all tests. After the pretreatment, the catalysts were tested at 260 °C, 1.5 MPa, H_2/CO = 0.67 and gas space velocities of either 3.3 l (STP)/g_{Fe} h (run SB-2537 with 100Fe/5Cu/6K/24SiO_2 catalyst) or 3.9–4.1 l (STP)/g_{Fe} h (runs SB-1327—silica-supported catalyst, SB-3007—alumina-supported catalyst, and SA-0946—100Fe/3Cu/4K/16SiO_2 catalyst). Tests SB-2537 and SA-0946 lasted 430 and 565 h, respectively. However, only the results obtained during the first 100 h on stream are described here.

3. Results and discussion

3.1. BET surface area measurements

The BET surface areas of calcined supports (air at 500 °C for 5 h) were 308 m^2/g (Davison Grade 952 silica support) and 195 m^2/g (Vista Catapal B alumina support—pseudo boehmite), whereas the corresponding pore volumes were 0.7 and 0.45 cm^3/g (Table 2). These values were determined by a standard BET multi-point method. The surface areas of supports before and after impregnation, determined by single-point BET method, are also listed in Table 2. Surface areas determined by the single-point method and standard multi-point method are in good agreement. Comparing the BET surface areas of supports, before and after the impregnation, it is observed that the surface area of both supports is reduced markedly after addition of iron, copper, and potassium. Decrease in the surface area of supports is possibly due to pore filling and/or pore blocking of mesopores during the impregnation step. Surface areas of calcined precipitated catalysts (260–310 m^2/g) are considerably higher than the corresponding surface areas of the supported catalysts (100–136 m^2/g).

Table 2
Textural properties of supported and precipitated iron catalysts

Catalyst or support	Surface area[a] (m^2/g)		Pore volume[a] (cm^3/g)
	Single-point	Multi-point	
Silica (Davison Grade 952)	252	308	0.70
Alumina (Vista Catapal B)	213	195	0.45
100Fe/5Cu/6K/139SiO_2	100	–	–
100Fe/5Cu/6K/139Al_2O_3	136	–	–
100Fe/3Cu/4K/16SiO_2	310	306	0.45
100Fe/5Cu/6K/24SiO_2	258	284	0.51

[a] Surface area and pore volume measurements of supports were made after calcination in air at 500 °C for 5 h. All catalysts were calcined in air at 300 °C for 5 h.

3.2. Temperature-programmed reduction (TPR)

Results from TPR measurements of the supported and precipitated catalysts in 5% H_2/95% N_2 are shown in Fig. 1 and Table 3. All four profiles show two distinct reduction peaks which are characteristic of the two-step reduction process of Fe_2O_3. It has been postulated that the first peak corresponds to reduction of Fe_2O_3 to Fe_3O_4 (magnetite), whereas the second broad peak corresponds to subsequent reduction of Fe_3O_4 to metallic iron [24–26]. Results show that the onset of reduction of Fe_2O_3 occurs earlier on the precipitated iron catalysts (first stage of the reduction process) and that this reduction process is more

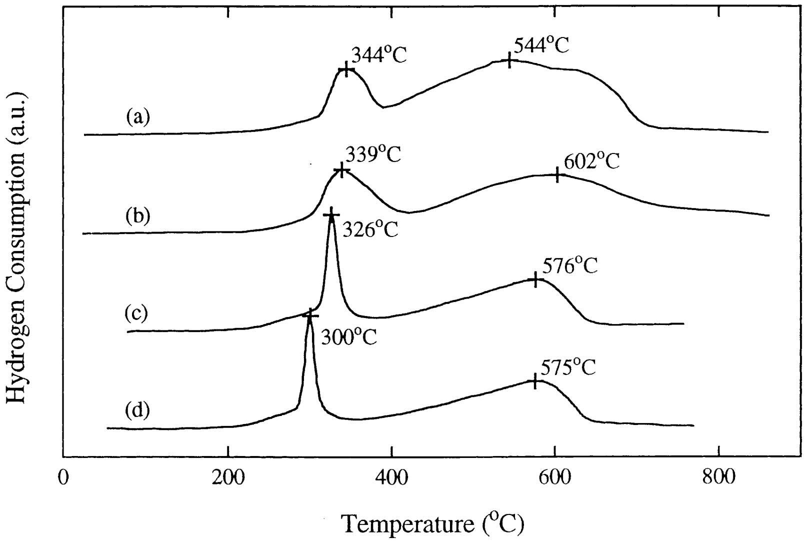

Fig. 1. TPR profiles of supported and precipitated iron catalysts: (a) silica-supported catalyst; (b) alumina-supported catalyst; (c) 100Fe/3Cu/4K/16SiO_2 catalyst; (d) 100Fe/5Cu/6K/24SiO_2 catalyst.

facile than on the supported catalysts (sharp narrow peaks for precipitated catalysts versus broad peaks for supported catalysts). The second stage of the reduction process is much slower on all four catalysts and the reduction is incomplete up to 800 °C in the temperature-programmed mode of reduction (Table 3).

Quantitative results from hydrogen uptake measurements during the TPR are summarized in Table 3, in the form of degree of reduction of iron (percentage of total iron reduced assuming that after calcination iron is in the form of hematite—Fe_2O_3). Also, listed in Table 3 are experimental ratios of the second peak area to the first peak area, and the corresponding theoretical uptakes corresponding to the reduction of magnetite (Fe_3O_4) to α-Fe (peak 2) compared to the total area of the first peak which represents the sum of areas (H_2 uptakes) for reduction of CuO to Cu and Fe_2O_3 to Fe_3O_4. Results listed in Table 3 represent average values from 3 to 5 measurements for each catalyst, and are reported in the form: average value ± S.D. These results show that the degree of reduction of iron after the first stage of the reduction process varies between 19 and 26%, which is considerably higher than the theoretical amount corresponding to the reduction of Fe_2O_3 to Fe_3O_4 (11.1% degree of reduction). The expected degree of reduction from Fe_2O_3 to FeO (wustite) is 33.3%. Also, experimental values of peak ratios of the second stage to the first stage of reduction are

Table 3
TPR results

Catalyst	Peak area ratio = area of the second peak/area of the first peak		Degree of iron reduction (%) RT to 800 °C	
	Theoretical value[a]	Experimental value	First stage	Total
100Fe/5Cu/6K/139SiO_2	6.33	3.6 ± 0.7	18.8 ± 2.2	84.8 ± 3.1
100Fe/5Cu/6K/139Al_2O_3	6.33	2.5 ± 0.2	23.0 ± 2.6	80.7 ± 8.0
100Fe/3Cu/4K/16SiO_2	6.90	2.5 ± 0.6	24.2 ± 1.2	82.0 ± 9.3
100Fe/5Cu/6K/24SiO_2	6.33	2.4 ± 0.4	25.6 ± 2.8	89.0 ± 6.4

[a] Calculated uptake ratio for ($Fe_3O_4 \rightarrow Fe$)/($Fe_2O_3 \rightarrow Fe_3O_4 + CuO \rightarrow Cu$).

significantly smaller than the corresponding theoretical values based on the assumed two stage reduction process consisting of: $Fe_2O_3 \rightarrow Fe_3O_4 + CuO \rightarrow Cu$ (first stage) and $Fe_3O_4 \rightarrow Fe$ (second stage). Smaller than expected area ratio is the result of two factors. The reduction of Fe_2O_3 during the first stage proceeds beyond Fe_3O_4, and the reduction during the second stage is incomplete (vide infra). Total degrees of iron reduction (up to 800 °C) obtained with all four catalysts are similar to each other, within experimental errors.

During isothermal reductions of 100Fe/3Cu/4K/16SiO_2 catalyst with H_2 at 240–280 °C in a slurry reactor, both Fe_3O_4 and α-Fe were positively identified by X-ray diffraction (XRD) and/or Mössbauer spectroscopy measurements, but wustite was not detected in partially reduced catalyst samples [27,28]. The same two phases (Fe_3O_4 and α-Fe) were identified in isothermal reduction studies with Ruhrchemie catalyst [29,30] in a fixed bed reactor (H_2 reductions at 220–280 °C). Ruhrchemie's catalyst composition (100Fe/5Cu/4.2K/25SiO_2) is very similar to that of 100Fe/5Cu/6K/24SiO_2 catalyst used in this study. Results from isothermal and TPR studies suggest that a small portion of iron oxide is reduced to α-Fe during the first stage of reduction (temperatures up to 350–400 °C), whereas most of Fe_3O_4 is reduced during the second stage of reduction in TPR mode. Identification of iron phases in samples after the first stage of reduction (e.g. by XRD and/or Mössbauer spectroscopy measurements) is needed to test validity of this hypothesis.

3.3. Isothermal reduction studies

Although, the TPR provides useful information on relative reducibility of different catalysts and energetics of the reduction process, it is not representative of reduction procedure employed in reaction studies. In the latter case, catalyst activation (pretreatment) is normally conducted under isothermal conditions with pure hydrogen, pure CO or synthesis gas. Isothermal reduction experiments in TGA unit are more representative of actual pretreatment conditions employed in the present study (except for differences in flow rate and partial pressure of the reductant gas). Initially, a catalyst sample was heated in flowing helium (40 cm^3/min) from room temperature to 280 °C, and then held at this temperature for 30 min to remove adsorbed moisture from the sample. The sample weight at the end of the heating period in helium is taken as a reference weight of the sample. After that, the flow is switched to either pure H_2 or pure CO, and the weight loss is recorded as a function of time. The observed weight loss is related to the degree of reduction of Fe_2O_3. For hydrogen reductions the degree of reduction is calculated by dividing the observed weight loss with the theoretical weight loss corresponding to the complete reduction of Fe_2O_3 to metallic Fe (the contribution from reduction of CuO to Cu is neglected, whereas the other oxides in a sample are assumed to remain in the oxide form). For CO reductions the degree of reduction cannot be calculated directly due to uncertainty in the final (and intermediate) form of iron compound and complications due to possible carbon deposition (see in the subsequent sections). Instead, the results are reported as the weight loss per amount of iron originally present in the sample, in order to account for different initial amounts of iron in different catalysts.

Results in Fig. 2 (hydrogen reductions) show that the rate of reduction is very rapid initially, and then gradually decreases with time. The reduction was incomplete after about 8 h at 280 °C. Final degree of reduction was approximately 80% for the two precipitated catalysts and the silica-supported catalyst, and only 25–30% for the two alumina-supported catalysts. The alumina-supported catalyst with a higher amount of potassium loading (9 parts of K/100 parts of iron by weight) had a lower degree of reduction than the catalyst containing 6 pbw of K/100 pbw of Fe. Reduction behavior of the silica-supported catalyst was very similar to that of the precipitated catalysts.

Reduction behavior of the supported and precipitated iron catalysts in CO at 280 °C is shown in Fig. 3. The initial rate of reduction (weight loss) was high for all catalysts, followed by gradual decrease with increase in time. Like in the case of hydrogen reductions, the reduction behavior of the two precipitated catalysts and that of the silica-supported catalyst was similar, whereas the two alumina-supported catalysts had a significantly lower degree of reduction. The reduction behavior of the two alumina-supported catalysts (containing different amounts of potassium) was similar. The alumina-supported catalyst containing 9 pbw of K exhibited continuous loss of weight with time, whereas the weight loss for the other four catalysts

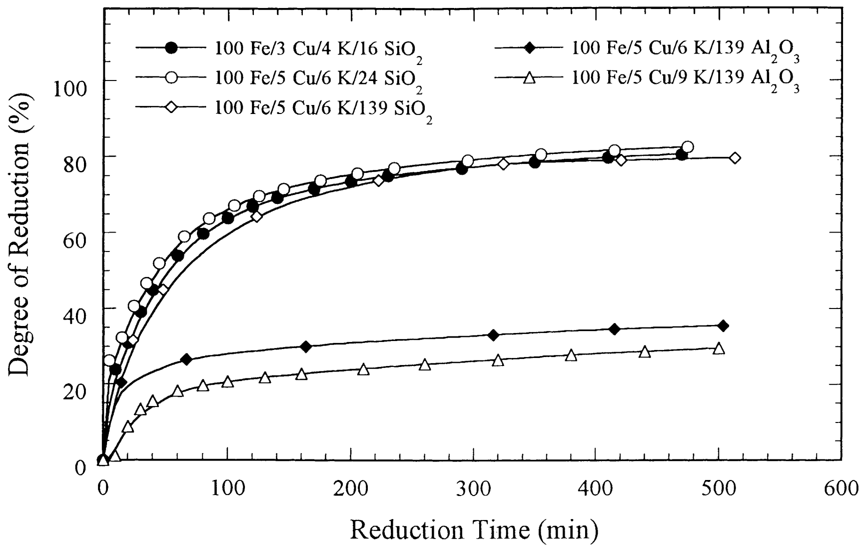

Fig. 2. Isothermal reduction behavior in hydrogen at 280 °C.

reached a minimum value after 150–200 min of reduction, followed by a slow increase in weight with time up to 500–540 min.

The observed changes in weight for all catalysts during the CO reduction are the net result of several competing reactions: (a) reduction of Fe_2O_3 to lower iron oxides, and possibly to metallic iron; (b) carbon deposition by Boudouard reaction ($2CO \rightarrow CO_2 + C$); and (c) carbide formation (i.e. carburization) directly from iron oxides or from the metallic iron. Theoretical weight losses for reduction of Fe_2O_3 to FeO, Fe, or iron carbides (χ-Fe_5C_2 or ε'-$Fe_{2.2}C$) are approximately: 10, 30 and 23.2–24%, correspondingly. The maximum experimental weight losses

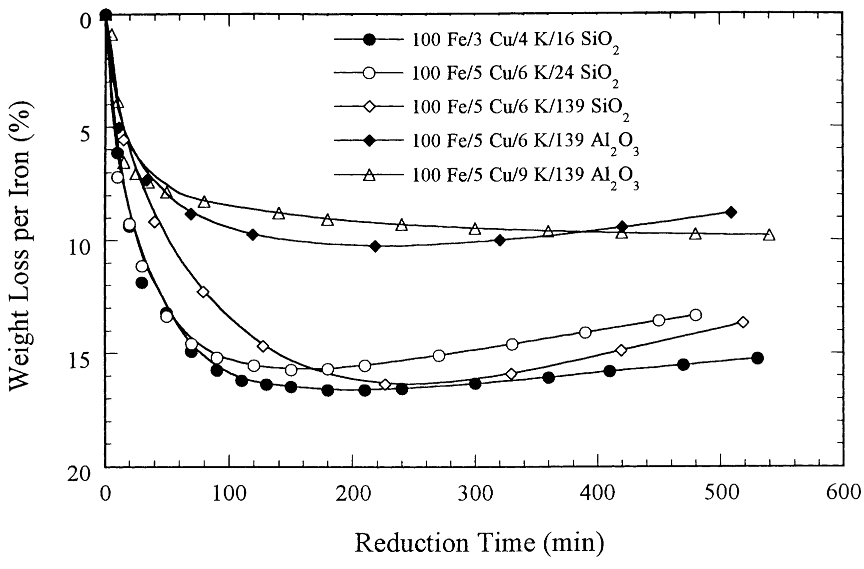

Fig. 3. Isothermal reduction behavior in CO at 280 °C.

were about 10% (alumina-supported catalysts) and 15–17% (precipitated catalysts and silica-supported catalyst). These results suggest that carburization is incomplete. A gradual increase in weight, observed in experiments with four of the five catalysts, after 150–200 min of reduction indicates that the carbon deposition is the dominant reaction, even though the carburization was not completed. Indirect evidence for carbon deposition during the CO reduction, via CO disproportiation reaction, was obtained in earlier studies with precipitated iron catalysts conducted in our laboratory [29,31] through measurement of the amount of CO_2 produced. Experimentally determined amounts of CO_2 produced during the CO reduction were in excess of the amounts required for complete Fe_2O_3 reduction to metallic iron followed by carbide formation.

The issue whether the formation of iron carbides occurs directly from iron oxides or through metallic iron has not been resolved by the present data. Although the maximum observed weight loss was considerably lower than that corresponding to Fe_2O_3 reduction to Fe, it is possible that only a small portion of iron oxides was completely reduced to metallic iron followed by carburization, whereas the bulk of iron is present in the form of oxides. An alternative explanation of the observed results is that iron carbides are formed directly from iron oxides, without formation of metallic iron as an intermediate step. Previous XRD and Mössbauer spectroscopy studies [27,29,31] of precipitated iron catalysts reduced in situ with CO at 280 °C for 8 h, in either fixed bed or slurry reactors, showed the existence of either iron carbide (χ-Fe_5C_2) only, or both iron carbide (χ-Fe_5C_2) and iron oxide (Fe_3O_4). Detailed characterization studies are required to determine whether carbide formation occurs through metallic Fe or directly from iron oxides.

3.4. Catalyst activity and stability

Changes in synthesis gas conversion with time-on-stream (TOS) for all four tests are shown in Fig. 4a. Catalyst activity in all tests first increased with time, and then either stabilized (precipitated catalysts) or began to decline with time (supported catalysts). The alumina-supported catalyst (run SB-3007) reached the maximum value of conversion at about 20 h on stream (45%) and then started to deactivate (35% conversion at 100 h). The loss of activity was less pronounced on the silica-supported catalyst, where conversion decreased from 67% at 20 h on stream to 60% at 100 h on stream.

Process conditions, including gas space velocities, in runs SB-1327, SB-3007 and SA-0946 were nearly the same and, therefore, measured values of syngas conversions are indicative of intrinsic catalyst activities in these three tests. Clearly, the alumina-supported catalyst was the least active, whereas the silica-supported catalyst was the most active up to about 25 h on stream. However, after 25 h on stream the precipitated 100Fe/3Cu/4K/16SiO_2 catalyst in run SA-0946 became the most active, and its activity continued to increase up to 80 h, whereas activity of the silica-supported catalyst started to decline with time. After 20 h on stream, syngas conversion in run SB-2537 (precipitated 100Fe/5Cu/6K/24SiO_2 catalyst) was higher than that of the silica-supported catalyst. However, the gas space velocity in run SB-2537 was lower (3.3 l/g_{Fe} h versus 4.0 l/g_{Fe} h in run SB-3007) and no definite conclusion can be drawn about intrinsic activities on the basis of syngas conversion data.

In order to obtain a quantitative comparison of catalyst activities, which accounts for differences in gas space velocities, a simple model was used to estimate values of an apparent reaction rate constant [9,23]. The apparent reaction rate constant was calculated assuming that F–T synthesis reaction is first-order with respect to hydrogen, and results are shown in Fig. 4b. At high syngas conversions (>60–70%) the F–T reaction rate becomes inhibited by water [1,17,23], and this type of analysis provides only an approximate value of the apparent rate constant. However, this approximation is still useful for relative comparison of catalyst activities [9,27]. As expected, the apparent rate constants for runs SB-1327, SB-3007 and SA-0946 follow the same trends as the corresponding values of conversion in Fig. 4a. The alumina-supported catalyst was the least active, and its apparent reaction rate constant was between 100 and 175 mmol (H_2 + CO)/g_{Fe} h MPa. Precipitated 100Fe/3Cu/4K/16SiO_2 catalyst was the most active with the apparent rate constant of about 400 mmol/g_{Fe} h MPa (after 70 h on stream). The latter catalyst is one of the most active iron F–T catalysts [9,11]. The activity of the silica-supported

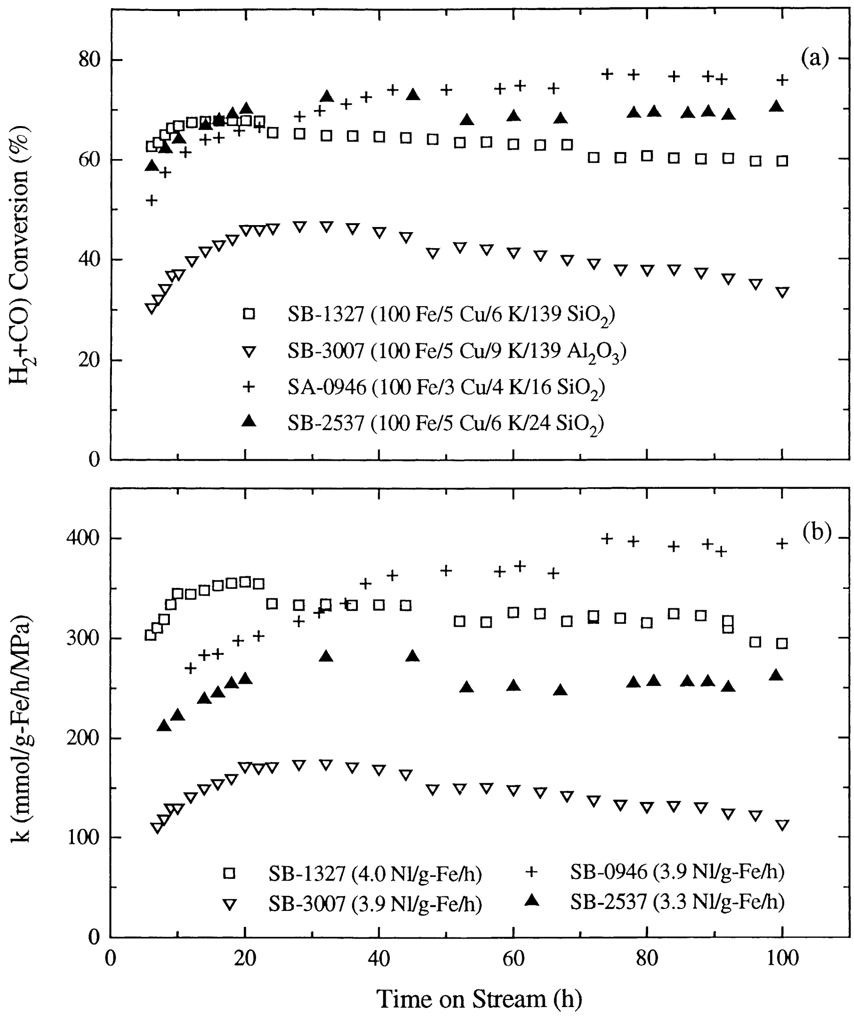

Fig. 4. Catalyst activity and stability with TOS at 260 °C, 1.5 MPa and $H_2/CO = 0.67$: (a) synthesis gas conversion; (b) apparent first-order reaction rate constant.

catalyst was higher than that of the precipitated 100Fe/5Cu/6K/24SiO_2 catalyst.

Catalyst activity of the supported catalysts correlates very well with their reduction behavior, i.e. the higher degree of reduction (under isothermal reduction conditions)results in higher catalyst activity. Also, the apparent rate constant of the silica-supported catalyst was similar to that of the precipitated catalysts, which is consistent with the observed similarity in their reduction behaviors. Initial activities of the two precipitated catalysts were similar to each other, which is also consistent with the observed similarity in their reduction behaviors. However, the activity of 100Fe/3Cu/4K/16SiO_2 catalyst continued to increase up to 80 h on stream and its steady state activity was significantly higher than that of 100Fe/5Cu/6K/24SiO_2 catalyst. This is ascribed to differences in the nature of iron carbide phases present and the crystallite particle size distribution, both of which vary with TOS[28,32].

O'Brien et al. [13] evaluated four supported catalysts (100Fe/6Cu/8.1K/250-260 support material) in a slurry reactor at 250 °C, 1.31 MPa, 3.1 l (STP)/g_{Fe} h and $H_2/CO = 0.7$. They found that alumina and magnesium aluminate supported catalysts were more active (higher CO conversion at the same process conditions) and stable than the silica and magnesium silicate supported catalysts. Reported activity trends with the silica and alumina-supported catalysts are opposite from the ones observed in the present study due to differences in support materials used, Fe loading, preparation method and/or activation procedures employed.

3.5. Gaseous hydrocarbon selectivities

Methane and gaseous hydrocarbon (C_2–C_4) selectivities shown in Fig. 5, expressed on mol% carbon atom basis, are defined as

$$S_{ij}\ (\%) = \frac{100(in_{ij})}{(n_{CO})_{in} - (n_{CO})_{out} - (n_{CO_2})_{out}}$$

where S_{ij} is the selectivity of hydrocarbon species j containing i carbon atoms, n_{ij} the molar flow of compound j in the gas phase, $(n_{CO})_{in}$ and $(n_{CO})_{out}$ are the molar flow rates of CO in and out of the reactor, and $(n_{CO_2})_{in}$ is the molar flow rate of carbon dioxide out of the reactor. This formula assumes that there is no carbon dioxide in the feed.

Silica-supported catalyst had the highest methane (6–7 mol%) and C_2–C_4 selectivities (26–28 mol%).

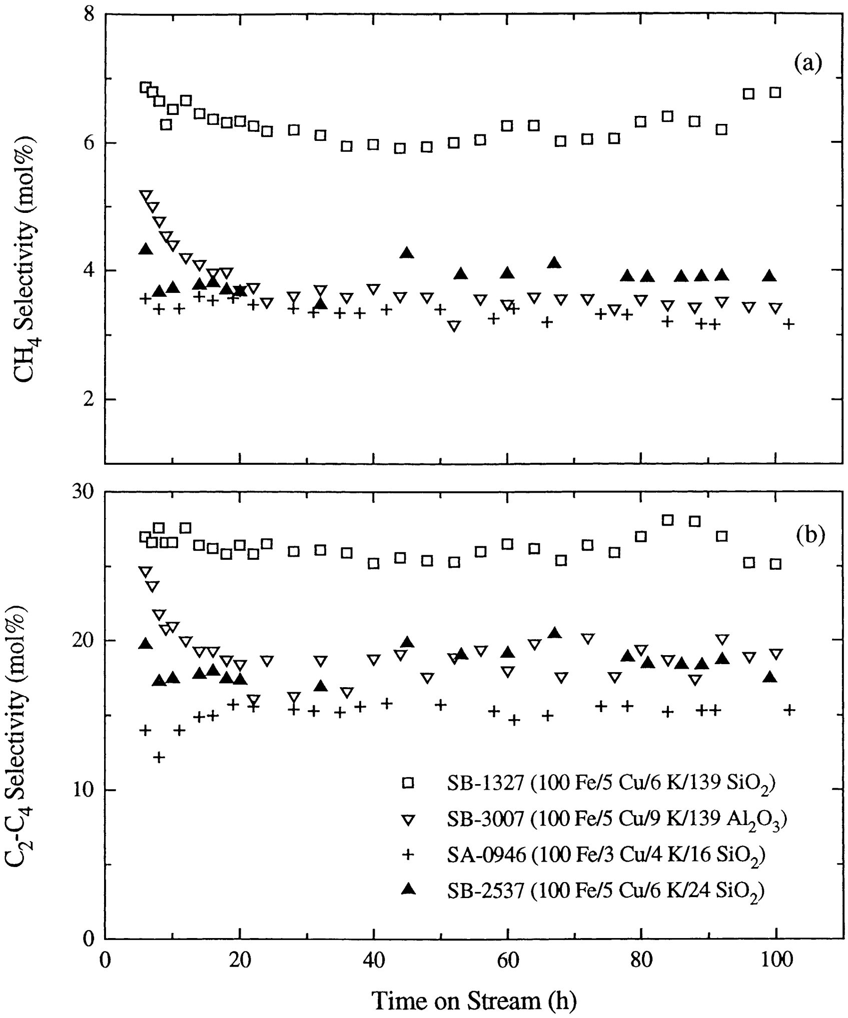

Fig. 5. Methane (a) and gaseous hydrocarbon selectivity (b) as a function of time. Process conditions are the same as in Fig. 4.

Methane and gaseous hydrocarbon selectivities on the alumina-supported catalyst were significantly lower and similar to those obtained in tests with precipitated catalysts. We ascribe these findings primarily to reduction in potassium promotion effectiveness on the silica-supported catalyst, which is caused by reaction between potassium and silica support [1] and/or by reduced contact between iron and potassium on a high surface area support [19]. Potassium addition is known to promote longer chain and more olefinic hydrocarbons [1,21,33]. The silica-supported catalyst, which has the same or greater amount of K relative to Fe as the two precipitated iron catalysts, exhibits selectivity characteristics typical of a catalyst with lower amount of potassium. On the other hand, promotional effect of potassium on the alumina-supported catalyst is similar to that of the precipitated catalysts, i.e. there are no indications of significant reductions of K promotional effect on hydrocarbon chain growth length. O'Brien et al. [13] also reported significantly lower methane selectivity on a alumina-supported catalyst in comparison to a silica-supported catalyst. These findings are contrary to Dry's reports [1,34], based on studies conducted at Sasol, that alumina supports are of "acidic" nature, and that the use of alumina support results in severe reduction of potassium promotional effectiveness, i.e. in poor catalyst performance (low activity and poor selectivity). Apparently, the observed differences in reduction of potassium promotional effectiveness on alumina-supported iron catalysts, were caused by use of different types of alumina materials (and/or impurities present), with widely varying degrees of "acidity". It appears that either support acidity is not so important factor in governing interaction between K, Fe and support as is surface area (alumina versus silica supports), or that aluminas used in the present study and O'Brien et al. study are less acidic than those used in earlier studies at Sasol [1,34].

Also, one may argue that the observed differences in gaseous selectivities, may have been caused either entirely or in part by differences in relative amounts of K on these two catalysts (more K on the alumina-supported catalyst) and/or differences in conversion levels in tests SB-1327 and SB-3007 (Fig. 4). We believe that these two effects have had only a minor effect on the observed results as explained below.

O'Brien et al. [13] reported that the alumina-supported catalyst has significantly lower methane yield than the silica-supported one (∼5% on the alumina support versus 8–10% on the silica support) at equal relative amounts of potassium (8 pbw of K/100 pbw of Fe). This supports our contention that significantly lower methane and C_2–C_4 hydrocarbon selectivities observed in our study on the alumina-supported catalyst, relative to the silica-supported one, are not caused primarily by higher level of K promotion. However, we do not imply that higher K loading on the alumina-supported catalyst did not have any effect on the observed hydrocarbon product distribution.

Next we discuss potential effect of different levels of conversion (∼35–40% in test SB-3007 with the alumina-supported catalyst versus ∼60–65% in test SB-1327 with the silica-supported catalyst) on hydrocarbon product distribution. On iron F–T catalysts the effects of conversion (or gas space velocity), (H_2/CO) feed ratio, and (H_2/CO) exit ratio are related to one another due to the water–gas shift reaction. Dry [1,34], summarizing results from Sasol's fixed bed and fluid bed reactor studies, stated that the most important controlling parameter of product selectivity in fixed bed reactors (precipitated iron catalysts) is the (H_2/CO) ratio within the reactor, whereas a combination of partial pressures of hydrogen, carbon monoxide and carbon dioxide has the most significant effect on selectivity in high temperature F–T synthesis in fluid bed reactors. Dictor and Bell [32] and Matsumoto and Satterfield [35] studied the effect of (H_2/CO) ratio on chain growth probability factor in stirred tank slurry reactors, and found that chain growth probability decreases (i.e. more low molecular weight products are formed) with increase in the reactor (H_2/CO) ratio, which is in agreements with findings from fixed bed reactor studies [1,33,34]. In the present study, the reactor (H_2/CO) ratio in runs with the supported catalysts (tests SB-1327 and SB-3007) was 0.64–0.70, and 1.0–1.2 in tests SB-2357 and SA-0946 with the precipitated catalysts. In tests with the supported catalysts, both catalysts were exposed to essentially the same gas environment, which should result in similar gaseous hydrocarbon selectivities in spite of differences in conversion levels. On the other hand, the conversions in the test of the silica-supported catalyst and the two precipitated catalysts were similar, and yet the silica-supported catalyst had much higher yield of methane and C_2–C_4 hydrocarbons. The reactor (H_2/CO) ratio in test SB-1327 with the silica-supported catalyst was lower than in tests with the precipitated catalysts, and this would tend to favor lower yields of gaseous hydrocarbon products on the silica-supported catalyst. Also, in our study [21] with precipitated iron catalysts of composition 100Fe/5Cu/4.2K/zSiO_2 ($z = 0$, 8, 24 and 100) it was found that conversion level and/or the exit (H_2/CO) ratio had very little effect on hydrocarbon product distribution (tests with the same catalyst at different gas space velocities). On the basis of these results we conclude that differences in conversion and/or the reactor (H_2/CO) ratio did not play a significant role in determining the gaseous hydrocarbon product distribution on catalysts evaluated in this study.

3.6. Olefin selectivity

Total olefin content or olefin selectivity (defined as $100 \times$ linear olefins/(linear olefins + n-paraffins)) as a function of carbon number in all four tests is shown in Fig. 6. Reported selectivities were calculated from gas chromatographic analysis of all products (gas phase, liquid phase and wax from the reactor) collected between 60 and 80 h on stream. Synthesis gas conversion in different tests, during this time period, varied between 40% (alumina-supported catalyst) and 76% (100Fe/3Cu/4K/16SiO_2 catalyst) as shown In Fig. 4. The observed carbon number dependence (maximum value at C_3–C_4) is typical for iron F–T catalysts, and has been attributed to secondary hydrogenation of 1-olefins [32,36] and/or diffusion enhanced 1-olefin readsorption [37,38]. Ethylene is more reactive than other low molecular weight olefins. Decrease in olefin content with increase in molecular weight has been ascribed to greater adsorptivity [33,36], higher solubility resulting in longer residence time in a slurry reactor [32,39], and/or lower diffusivities [37,38] of high molecular weight hydrocarbons.

The highest olefin selectivity was obtained in tests of the alumina-supported catalyst and precipitated 100Fe/3Cu/4K/16SiO_2 catalyst, whereas the silica-supported catalyst had the lowest olefin selectivity. High olefin selectivity in run SB-3007 is due to low syngas conversion (low conversion favors primary reactions), high potassium loading of the catalyst and absence of significant interaction between the alumina support and potassium. Potassium promotion suppresses secondary reactions, and increases olefinicity of the products [1,33,39–42]. Silica-supported catalyst has the same (or higher) potassium loading as the two precipitated catalysts, but its olefin selectivity is lower. Also, the reactor (H_2/CO) ratio in test SB-1327

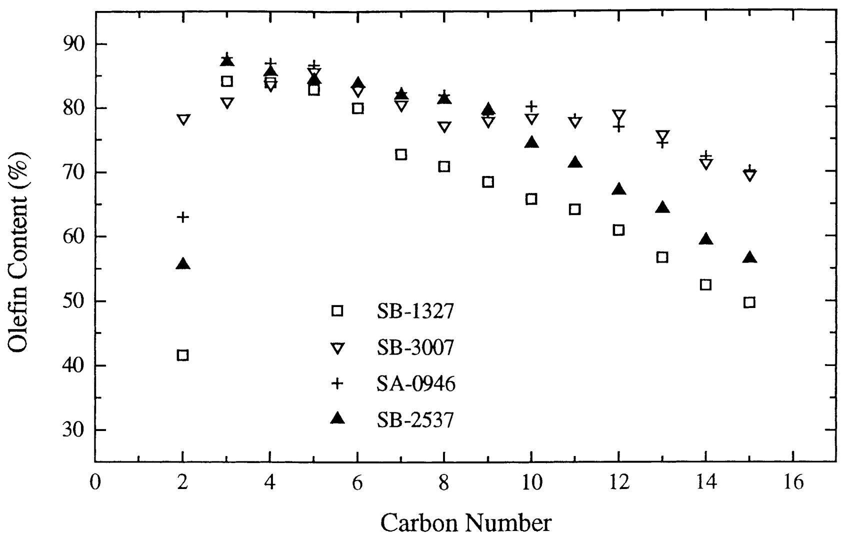

Fig. 6. Comparison of total olefin selectivity (TOS = 60–80 h; syngas conversion = 40% for the alumina-supported catalyst; 60–75% for the other catalysts).

was lower than that in tests SB-2357 and SA-0946 with the precipitated catalysts, which should favor higher olefin selectivity on the silica-supported catalyst. Again, these results are ascribed to interaction between potassium and the silica support (i.e. reduction in the effectiveness of K promotion). Olefin selectivities on the two precipitated catalysts also reflect reduction in the effectiveness of K promotion with increasing amount of silica binder. The same type of behavior, i.e. decrease in olefinicity with increasing amount of silica, was observed in fixed bed reactor studies with precipitated iron F–T catalysts [1,21,43].

3.7. Concluding remarks

Reduction of iron oxide in temperature-programmed mode, with both supported and precipitated catalysts, occurs in two-steps (stages). Hydrogen consumption during the first stage of reduction (~250–400 °C temperature range) was significantly higher than the theoretical one corresponding to the reduction of Fe_2O_3 to Fe_3O_4 for all four catalysts. A plausible explanation for these results is that a small portion of iron oxide was reduced to metallic iron during the first stage of reduction. This hypothesis needs to be verified by XRD and/or Mössbauer spectroscopy measurements of iron phases present in a sample after the first stage of reduction. The majority of magnetite is reduced slowly to metallic iron during the second stage of reduction. Reduction rate of the two supported catalysts was slower in comparison to precipitated iron catalysts, but the calculated final degrees of iron reduction (81–88%) in TPR mode were similar for all catalysts.

Isothermal reductions at 280 °C in pure hydrogen and CO, revealed that alumina-supported catalysts have significantly lower degree of reduction than the silica-supported and the two precipitated catalysts. Reduction behavior (in H_2 and CO) of the silica-supported catalyst was similar to that of the precipitated catalysts. Reduction process with CO as a reductant, is complicated due to the presence of several simultaneous reactions: removal of oxygen from Fe_2O_3 (formation of lower iron oxides and possibly metallic iron), carburization of iron oxides and/or metallic iron (formation of iron carbides), and CO disproportiation resulting in carbon deposition. Gradual increase in sample weight after 150–200 min of CO reduction, observed with both supported and precipitated catalysts, has been ascribed to carbon deposition on the catalyst.

Contrary to previous reports [1,13,34] that activity of supported iron catalysts is lower than that of precipitated iron catalysts, we found that initial activity (per gram Fe basis) of the silica-supported catalyst was 20–40% higher than that of the two most active precipitated iron catalysts synthesized at TAMU. After 100 h on stream the silica-supported catalyst was still more active than one of the two precipitated catalysts. Alumina-supported catalyst was the least active (~50% less active than the silica-supported catalyst) and its deactivation rate was the highest of all four catalysts tested. Low activity of the alumina-supported catalyst is ascribed primarily to its lower degree of reduction. In addition to the degree of reduction, catalyst activity is determined by the nature of iron phases present and their crystallite particle size.

The silica-supported catalyst had lower olefin content and higher methane and C_2–C_4 hydrocarbon selectivities than the two precipitated catalysts and the alumina-supported catalyst. Hydrocarbon product distribution (including the total olefin content) of the alumina-supported catalyst was similar to that of the precipitated catalysts. The observed differences in selectivities have been interpreted in terms of reduction of potassium promotion effectiveness, which is caused by interactions between potassium and supports, binders and/or impurities present. These interactions were much stronger with a high surface area silica support (Davison Grade 952), than with the alumina support (Vista Catapal B), which is not consistent with expected higher degree of "acidity" of alumina relative to silica [1,34,44].

Results obtained in this study with supported catalysts are encouraging, however, further improvements in their performance are needed. The silica-supported catalyst had activity comparable to that of two highly active precipitated catalysts, but its selectivity was inferior. On the other hand, the alumina-supported catalysts had low methane selectivity characteristic of a high-alpha catalyst, but its activity was low in comparison to highly active precipitated iron F–T catalysts. Also, the stability of supported catalysts needs to be improved. Further studies with the supported iron catalysts are in progress in our laboratory, focusing on the use of different supports, preparation methods and/or promoter levels to improve the catalyst performance.

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